Motor Subtype Progression in Parkinson's Disease

Progress report: work completed after the midterm presentation

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Abstract

At the midterm we showed that a patient's recorded motor subtype changes often and noisily, and that a hidden Markov model predicts the next recorded subtype of held-out patients with 75.4% accuracy. This report covers the work done since then. We forecast the continuous tremor/PIGD ratio with Gaussian process regression, modelled the tremor and PIGD scores separately, tested whether the rate of progression changes over follow-up, checked how stable the hidden Markov model is, and asked whether patients fall into distinct trajectory groups. The main findings are: flexible models forecast continuous scores better than simple rules, but this does not translate into better prediction of the next recorded subtype; PIGD worsens faster than tremor, and the difference grows gradually over follow-up rather than at a clear transition point; the three-state hidden Markov model is stable, while larger models add nothing; and the clustering question remains inconclusive, because we could not build a sufficiently realistic comparison population without groups.

1 Introduction and context

Our project studies how the motor presentation of Parkinson's disease changes across repeated clinical visits, using the Parkinson's Progression Markers Initiative (PPMI) cohort [1]. At each visit a patient is classified as tremor-dominant (TD), postural instability and gait difficulty (PIGD) or Indeterminate, from the ratio of the mean of 11 tremor items to the mean of 5 gait and balance items of the MDS-UPDRS rating scale [2, 3]. A ratio of at least 1.15 is TD, at most 0.90 is PIGD, and values in between are Indeterminate.

The analysable cohort contains 934 patients with sporadic Parkinson's disease and at least two labelled visits, with 4,295 labelled visits in total. All results in this report use this cohort or, where stated, a subset of it.

At the midterm we reported three things that shaped the later work:

- The recorded subtype is noisy. About a third (35.2%) of observed subtype changes reverse at the next visit, and a patient's own ratio varies far more between visits than the width of the Indeterminate band.
- A hidden Markov model (HMM), which treats the true motor state as hidden and the recorded label as a noisy reading of it, predicted the next recorded subtype of held-out patients with 75.4% accuracy (3,361 predictions), slightly better than simple chains.
- A five-state version of the HMM converged in only 404 of 934 folds and gave different answers from different starting points.

We then proposed four directions: model tremor and PIGD separately rather than through their ratio; test whether the rate of progression is constant; ask whether patients fall into distinct

progression patterns; and understand when the HMM fails. This report describes what we did in each direction, and what we learned.

2 Objectives after the midterm

The work after the midterm was guided by five questions.

1. **Can continuous scores be forecast better than by simple rules?** The HMM predicts a category. We wanted to know whether forecasting the continuous scores behind the label is more informative.
2. **Is the ratio the right quantity to model?** The ratio hides severity (2/4 and 10/20 give the same ratio) and is undefined when the PIGD score is zero, which happens at 692 of the 4,295 labelled visits.
3. **Does the rate of progression change over follow-up?** Every earlier model assumed one constant rate.
4. **How stable is the HMM, and why does it never predict Indeterminate?**
5. **Do patient trajectories form distinct, reproducible groups?** If so, they could be candidate progression phenotypes.

Running through all five is a sixth question: how much of the variation we see is caused by measurement noise and by the irregular follow-up of patients, rather than by the disease itself.

3 How the work was evaluated

All prediction results in this report come from patients the model has not seen. We use *leave-one-patient-out* validation: each patient is removed in turn, the model is fitted on the remaining patients, and the removed patient's visits are predicted one at a time, using only that patient's earlier visits. This is the same protocol used for the HMM at the midterm.

Every model is compared with simple *baselines*, which are rules that need no modelling:

- **Persistence**: the next value (or label) equals the last observed one. Because labels change slowly on average, this is a strong baseline.
- **Patient mean**: the next value equals the patient's own average so far.

We report the *mean absolute error* (MAE), the average size of a forecast error, and the *continuous ranked probability score* (CRPS), which judges the whole predicted distribution, not only its centre; for both, lower is better [5]. For subtype predictions we report accuracy and *balanced accuracy*, the average of the accuracies within each of the three classes, which is fairer when Indeterminate is only about one visit in ten. Intervals in square brackets are 95% intervals from resampling patients (a patient bootstrap); when two models are compared, both are evaluated on the same resampled patients.

Two cohorts appear below. Models of the ratio can only use visits where the ratio is defined: 821 patients and 2,707 predicted visits. Models of the tremor and PIGD scores use all 934 patients and 3,361 predicted visits, which is exactly the HMM's set. Results from the two cohorts are never compared directly.

Most of the work was re-run on the college computing cluster. Before any new experiment, we confirmed that the data panel rebuilt on the cluster is identical to the one used for the midterm, and that earlier predictions were reproduced (to within 1.4×10^{-4}).

4 Work completed

4.1 Forecasting the continuous ratio with Gaussian process regression

Why. The HMM predicts a category and the midterm growth model assumed a straight-line trend. We wanted a flexible forecast of the continuous ratio for each patient, with an honest statement of its uncertainty.

What we did. Gaussian process regression (GPR) [4] predicts a patient's next value from their earlier visits, using a population trend plus a smooth, patient-specific deviation from it. How the deviation behaves over time is set by a *kernel*. We compared a linear-trend kernel (K0, which reproduces the midterm growth model) with three flexible kernels (K1–K3), on the 821-patient ratio cohort.

What we found. The flexible kernels forecast the ratio better than both the linear-trend model and the baselines (Figure 1). The MAE of K1 was 0.972 against 1.043 for persistence, a difference of −0.071 [−0.098, −0.045], and −0.033 [−0.047, −0.021] against K0. The three flexible kernels were practically identical (K3 minus K1: −0.0005 [−0.0015, +0.0005]). The same ordering held under three alternative ways of excluding patients with extreme ratios.

However, the better ratio forecasts did not give better subtype predictions. On the same 2,707 visits, the subtype derived from K1 was correct 72.4% of the time, against 73.3% for persistence, and the GP never predicted Indeterminate. About 13% of each predicted distribution lay below zero, a value the ratio cannot take. Over longer horizons the GP's advantage over persistence looked larger (MAE 0.976 against 1.605 for forecasts more than five years ahead, 158 forecasts from 100 patients), but these long horizons come only from patients who stayed in the study for many years, and we did not find a clear widening within three years.

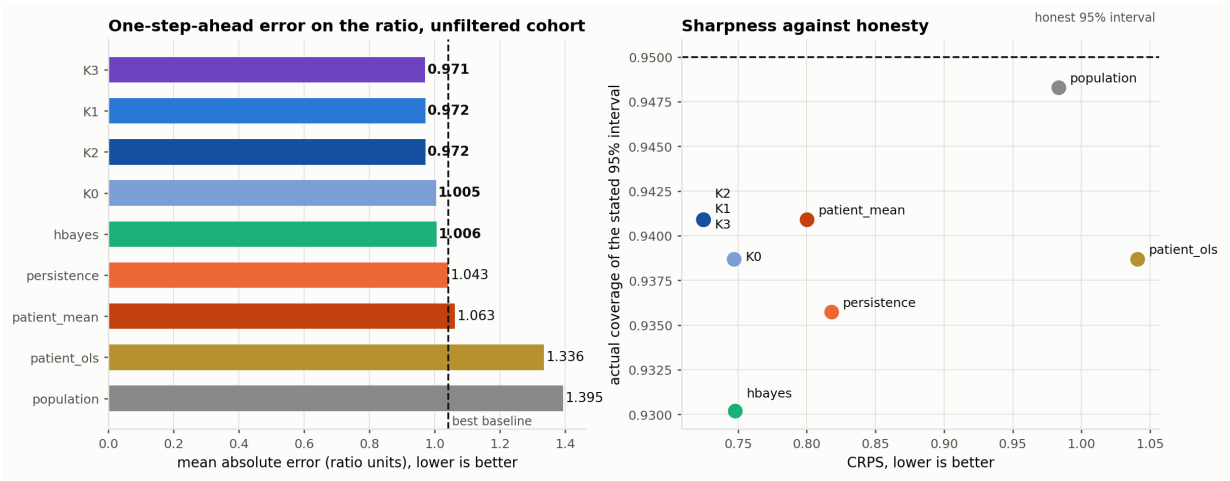


Figure 1: Held-out one-step-ahead forecasts of the TD/PIGD ratio (821 patients, 2,707 forecasts). Left: mean absolute error; the dashed line marks the best simple baseline (persistence). Right: CRPS (lower is better) against the share of observations falling inside each model's stated 95% interval (0.95 would be ideal). "hbayes" is the midterm growth model and "K0" its Gaussian process equivalent.

What we learned. Two things. First, flexible forecasting of the continuous score is possible and beats the simple rules. Second, the ratio is a poor quantity to forecast: it is undefined for many visits, it cannot be negative but a Gaussian model allows it to be, and better continuous forecasts do not help with the subtype label. This led directly to modelling the two scores separately.

4.2 Modelling tremor and PIGD scores separately

Why. Modelling the tremor and PIGD scores themselves keeps severity, uses every labelled visit (including the 692 with a PIGD score of zero), and places the work on exactly the same 3,361 predictions as the HMM, so the two can finally be compared fairly.

What we did. We compared, on identical held-out folds and visits: persistence and patient mean; a joint linear mixed model (LMM, a regression with a population trend and a patient-specific offset for each score); two separate Gaussian processes, one per score; and a joint (multi-output) Gaussian process, which can share information between the two scores. The joint model was fitted both with and without a cross-score link, so that the link could be tested on its own. All four kernels (K0–K3) were evaluated held out; earlier, the slowest two had only been fitted to the whole cohort because of computing time.

What we found.

- **Trends.** Both scores worsen, PIGD about 2.8 times faster than tremor: +0.066 per year [0.059, 0.073] against +0.023 [0.017, 0.030] (linear model, patient bootstrap). The ratio's decline is therefore driven by the PIGD score rising, not by tremor improving.
- **Joint modelling gives no benefit.** In 26 of 28 tests of the cross-score link the interval included zero. The two exceptions were differences of about 0.0005 against a typical error of 0.24. The estimated correlation between the two scores was close to zero.
- **Flexible kernels help, but only K2 and K3.** The two kernels with a linear trend plus a smooth deviation beat the linear-trend model on every probabilistic score for both outcomes (for example PIGD CRPS −0.0035 [−0.0053, −0.0018] for K3). K1 did not. An earlier conclusion that flexible kernels do not help on the subscores was based on K1 alone and is corrected by this result.
- **Against persistence the answer depends on the measure.** Persistence still has the smaller average error for PIGD (K3 minus persistence: +0.009 [+0.005, +0.013]), but the Gaussian process is better when the whole predicted distribution is scored (PIGD CRPS −0.013 [−0.015, −0.010]). This second result depends on the uncertainty we assigned to persistence.
- **The Gaussian process beats the LMM** on both average error and CRPS, for both scores (for example PIGD MAE −0.022 [−0.027, −0.017] for K1).

For the next recorded subtype (Table 1), the Gaussian process and the HMM were equally accurate (difference +0.002 [−0.005, +0.009]), and the Gaussian process gave better-calibrated probabilities (lower log loss: −0.020 [−0.032, −0.007]). The comparison with simply repeating the last label is more revealing. When each model predicts its most probable class, both gain about three points of accuracy over repetition, but they do so by never predicting Indeterminate, and their balanced accuracy is lower. When the decision rule is adjusted for how common each class is, the HMM is the only model whose balanced accuracy exceeds simple repetition (+0.019 [+0.007, +0.031]).

Table 1: Next recorded subtype, held-out (934 patients, 3,361 predictions). "Most probable" predicts the class with the highest probability; "adjusted" divides each probability by that class's frequency in the training patients.

Model	Decision rule	Accuracy	Balanced accuracy	Indeterminate recall
Repeat last label	–	0.723	0.548	0.193
HMM	most probable	0.754	0.526	0.000
HMM	adjusted	0.711	0.568	0.247
Gaussian process (K1)	most probable	0.756	0.532	0.000
Gaussian process (K1)	adjusted	0.650	0.554	0.220

What we learned. Modelling the scores separately is the better representation: it uses all visits, shows that PIGD drives the change, and allows a fair comparison with the HMM. It does not give better prediction of the next recorded subtype. No model in this work clearly beats repeating the last label on balanced accuracy except the HMM with an adjusted decision rule.

4.3 Does the rate of progression change over follow-up?

Why. All earlier models assumed a single constant rate. The midterm listed a changing rate as a hypothesis to test.

What we did. We fitted the joint mixed model with a rate that is allowed to change at fixed points (2 and 5 years of follow-up, chosen before fitting). We then checked this result in several ways: intervals from resampling patients instead of from the model's own formula; a test for a change at an unknown time, calibrated by simulating data with a constant rate; alternative smooth shapes; adjustment for treatment status and disease duration; and the effect of patients leaving the study.

What we found. The rate of change is not constant. PIGD worsens faster later in follow-up, while tremor slows (Figure 2). The PIGD rate after five years minus the rate in the first two years was +0.035 [+0.014, +0.056]; for tremor, −0.020 [−0.037, −0.002]. The calibrated test rejected a constant rate ($p \leq 0.005$, the smallest value possible with 200 simulations).

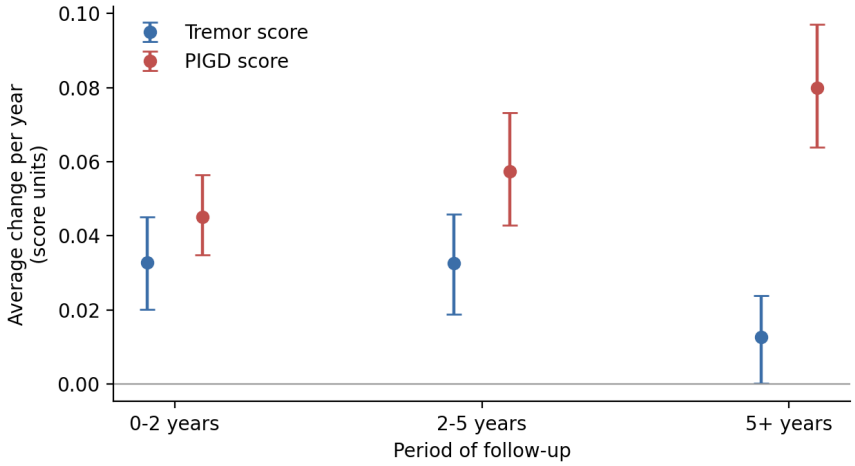


Figure 2: Average yearly change in the tremor and PIGD scores in three periods of follow-up, with 95% intervals from 1,000 patient-bootstrap refits (934 patients). The rates are population averages from a model with fixed change points at 2 and 5 years.

Several checks limit how this can be interpreted:

- The model's own intervals for the late period were 1.7–2.2 times too narrow; the figure shows the corrected ones.
- There is no identifiable transition point. The best single change point was 4.75 years, but across resampled datasets it ranged from 3 to 9.75 years (10th to 90th percentile), and a smooth curve fitted the data better than any model with change points.
- Follow-up time cannot be separated from disease duration. Patients joined PPMI within a few years of diagnosis, and after adjusting for disease duration the change at five years could no longer be estimated.
- Treatment starts in the same period: the share of examinations from treated patients rose from 0.10 in the first year to 0.54 in the second and about 0.95 after five years.
- Patients with higher PIGD scores were more likely to leave the study (log-odds +0.52 per point [+0.36, +0.71]). Even so, among the 247 patients followed for at least six years, PIGD still worsened faster late than early (+0.044 [+0.017, +0.072]).

Allowing the rate to change improved one-visit-ahead prediction only very slightly (PIGD CRPS −0.0010 [−0.0017, −0.0002]; no change for tremor).

What we learned. The divergence between tremor and PIGD is real and gradual in this cohort. It does not show a biological transition at five years, and with this study design we cannot say whether it belongs to time in the study, disease duration or treatment.

4.4 Stability of the hidden Markov model and the Indeterminate class

Why. At the midterm the five-state HMM gave different answers from different starting points, and no HMM ever predicted Indeterminate. We wanted to know whether the reported 75.4% rests on a stable fit, and why Indeterminate is never chosen.

What we did. We refitted several HMM versions (three states with different constraints on how reliably each state is recorded, four states and five states) from 40 widely spread starting points each. We then evaluated all of them on held-out patients in ten groups, choosing the version on the training patients only, so that the reported performance is not biased by the choice.

What we found.

- The three-state HMM used at the midterm reaches the same solution from all 40 starting points; its results do not depend on how it is started.
- Without its recording-reliability constraint, the three-state model reaches the same fit with very different parameters, so its parameters cannot be interpreted, although its predictions can be used.
- The five-state model stops at its evaluation limit from most starting points; with a larger budget it reaches 33 different solutions from 40 starts. It does not predict better than the three-state model.
- When the version is chosen on training patients only, held-out accuracy is 0.754 [0.735, 0.774], the same as reported at the midterm.
- No version ever gives Indeterminate more than about 0.4 probability, so a most-probable-class rule can never choose it. The probabilities do carry some information (they rank true Indeterminate visits above others 62% of the time). With the adjusted decision rule, Indeterminate recall rises to 0.25 and balanced accuracy from 0.526 to 0.571, at a cost in overall accuracy (0.712).

What we learned. The midterm HMM result is stable and was not inflated by model choice. Extra hidden states add nothing. The Indeterminate class is never predicted because of the decision rule combined with weak information, so overall accuracy alone describes these models poorly; balanced accuracy and probability scores should be reported with it.

5 Summary of important results

Table 2: Main results after the midterm. All prediction results are held-out.

Question	Approach	Main finding	What it means
Forecast the continuous ratio better?	Gaussian process regression, 821 patients	Beats persistence on MAE (-0.071 [-0.098, -0.045]); subtype accuracy 0.724 vs 0.733 for persistence	Better continuous forecasts, no better labels; the ratio is a poor target
Model tremor and PIGD separately?	LMM, separate and joint Gaussian processes, 934 patients	PIGD rises 2.8× faster than tremor; joint modelling adds nothing; K2/K3 beat the linear trend	Better representation, not better subtype prediction
Beat simple repetition on subtype?	GP and HMM vs repeat last label	Accuracy equal between GP and HMM; neither beats repetition on balanced accuracy except the HMM with an adjusted rule	Next-label prediction is close to its limit with these data
Constant rate of progression?	Piecewise mixed model plus checks	Rate not constant ($p \leq 0.005$); gradual, no identifiable change point	Real divergence, but not a demonstrated disease-stage transition
Is the HMM stable?	40 starts, nested held-out evaluation	Three-state model stable; held-out 0.754 [0.735, 0.774]; extra states add nothing	The midterm result stands
Distinct trajectory groups?	Reliability audit, simulated comparison populations	Inconclusive (Section 6)	No phenotype claim can be made

6 The clustering investigation and its conclusion

Motivation. We wanted to know whether patients form distinct trajectory groups, such as slow and fast progressors, or whether apparent groups arise from continuous differences between patients, from measurement noise and from how often patients were seen.

Are individual trajectories reliable? We first checked whether each patient's rate of change can be estimated reliably enough to group patients. Mostly it cannot. Removing a single visit reversed the direction of the estimated slope for 369 of 762 patients with three or more visits (tremor) and for 317 of 762 (PIGD). The median reliability of the patient-specific slope was 0.27 for tremor and 0.53 for PIGD, below the level usually needed to classify individuals. When each patient's last visit was removed, the agreement between the new and original two-group clustering (adjusted Rand index, 1 = identical, 0 = chance) was only 0.46.

Why a comparison simulation was needed. A clustering method always finds groups, even in data that has none. The fair question is therefore whether the real data shows stronger or more stable groups than data simulated from a population with *no* deliberately created groups

but with the same visit schedules, the same score scale and the same measurement noise. If such a no-groups population produces the same signals, the signals are not evidence of groups.

First comparison. Against 200 simulated no-groups populations, the real data separated into groups much more strongly than expected (a model selection score, ΔBIC, of 513 against a 95th percentile of 117), but the groups were no more stable when a visit was removed (agreement 0.750 against a 95th percentile of 0.771). Our pre-specified rule required both, so clustering was not supported. A later check showed that the simulated populations were themselves not realistic: the real patients' estimated PIGD slopes are much more skewed (skewness 2.17, against about 0.4–0.6 in the simulations), and the simulations produced too many PIGD scores of zero.

Final investigation. We therefore tried to build a more realistic no-groups population. Before each attempt we fixed the features it had to reproduce and the allowed tolerances. If any feature failed, the clustering comparison was not run. Table 3 summarises the attempts.

Table 3: Attempts to build a realistic population without groups. Each row was a pre-specified check of whether the simulated data reproduces the real data; no clustering comparison was run after any failure.

Model of the patients and the scores	What it reproduced	What it failed
Skewed, heavy-tailed patient differences on the original score model	Most shape targets	PIGD-slope skewness (1.73 vs 2.17); too many zero scores
Three models of the PIGD score (constant noise, noise growing with severity, beta-binomial)	Part of the PIGD-slope skew (beta-binomial)	Zero scores (about 24% vs 16%) and how patients move to zero
Item-level model of the five PIGD items (graded response [6])	Zero frequency, item response rates; 14 of 18 targets	How often patients stay at or move to zero; PIGD-slope shape
Item-level model plus a slowly varying within-patient deviation	14 of 18 targets	The same four targets; the added part was poorly determined by the data

Two findings came out of this work. First, part of the unusual shape of the real PIGD slopes comes from the score scale itself: with patient differences that are not skewed at all, a realistic item-level model already produced a PIGD-slope skewness of 1.72, against 0.50 for the original simulation. Second, real patients stay at or leave a PIGD score of zero more persistently than any model we tried allows. The last model was meant to capture this persistence, but it fitted a deviation that changes over about 13 years, which over the available follow-up behaves like a fixed patient offset and was not well determined by the data.

Conclusion.

- The clustering investigation was closed as inconclusive.
- An adequately calibrated no-groups comparison model was not obtained.
- The clustering comparison was not run after the pre-specified checks failed.
- The results do not establish that distinct trajectory groups exist.
- The results also do not establish that distinct groups do not exist.
- Large differences between patients, or unstable group assignments, are not by themselves evidence of distinct clinical phenotypes.

7 Overall discussion

Observed directly in the data. The recorded subtype fluctuates much more than the underlying disease is likely to. The PIGD score is a sum of five items that behave very differently: the gait item is scored above zero at 75% of visits, freezing of gait at 6%. Scores of zero are common early on, are not permanent, and become rarer with time. Variation between visits grows with severity. Follow-up is irregular, and patients with worse gait were more likely to leave the study.

Supported, but dependent on modelling choices. PIGD worsens faster than tremor, and the gap widens over follow-up. Flexible forecasts of the continuous scores beat simple rules on the measures we used, partly because of the uncertainty we assigned to the simple rules. The HMM's estimates of the hidden state are model outputs, not observed clinical truth. Part of the skew in the patient-specific PIGD slopes is produced by the score scale.

Unresolved. Whether patients fall into distinct trajectory groups; why patients near a PIGD score of zero behave more persistently than our models allow (possible explanations include how individual items are scored near the floor, changes in item behaviour with treatment or time, or dependence between successive visits, none of which we have tested); and whether the late acceleration of PIGD reflects disease stage, treatment or who remained in the study.

Limitations. All results come from one cohort (PPMI). The prediction target is the label produced by our own classification rule, not an independent clinical diagnosis, so held-out accuracy measures agreement with a noisy record. Our results concern the next visit; they do not show that an individual's long-term trajectory can be predicted. Several comparisons with persistence depend on the uncertainty we assigned to it. Patients who left the study early are under-represented in the later years.

These limitations are not all failures. Several of them, such as the noisy label, the effect of the score scale and the confounding of follow-up time with disease duration, are things we now understand and can state precisely, and that earlier analyses did not take into account.

8 Conclusion

After the midterm we asked whether continuous forecasting, modelling the two scores separately, a changing progression rate or trajectory groups could add to the HMM picture. Continuous forecasting works and beats simple rules on continuous measures, but it does not predict the next recorded subtype better; next-label prediction is close to what these data allow, and only the HMM with an adjusted decision rule improves balanced accuracy over repeating the last label. Modelling tremor and PIGD separately is the better representation and shows that PIGD drives the change, with a gradual acceleration that we cannot attribute to a specific cause. The three-state HMM presented at the midterm is stable and its accuracy was not inflated. The question of distinct trajectory groups remains open, because a realistic comparison population could not be built.

We have not shown that any model can predict an individual patient's long-term trajectory, and none of the models has been validated for clinical use.

9 Future work

Completed. All analyses described above are complete, and their results and settings are recorded in the project repository. The clustering investigation is closed; we do not plan to restart it.

Proposed.

- Report balanced accuracy and probability scores alongside accuracy in every subtype result, and choose any decision rule on training data.
- Explore the Indeterminate class further, for example which visits receive the highest Indeterminate probability, and whether a sustained Indeterminate period means something different from a single visit.
- Use the tremor and PIGD scores, and possibly the individual items, instead of the ratio wherever the label is not essential.
- Study how patients move to and from a PIGD score of zero, as a descriptive question about measurement, without building another clustering comparison.
- Update the earlier explanatory report where its conclusions have been corrected, and consolidate everything for the final BTP submission.

References

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A Models and evaluation, in brief

Term	Meaning in this report
Hidden Markov model (HMM)	The true motor state is hidden and changes over time; the recorded label is a noisy reading of it.
Gaussian process regression	Forecasts a patient's next value from their earlier visits: population trend plus a smooth patient-specific deviation. K0 = linear trend; K1 = smooth deviation; K2, K3 = linear trend plus smooth deviation.
Linear mixed model (LMM)	Regression with a population trend and a patient-specific offset.
Leave-one-patient-out	Each patient is held out in turn; the model never sees the patient it predicts, and only earlier visits are used.
Persistence	Predicts that the next value or label equals the last one.
MAE	Average absolute forecast error.
CRPS	Score of the whole predicted distribution; lower is better.
Balanced accuracy	Average of the accuracies within each class.
ΔBIC	How much better a model with groups fits than one without; larger means stronger apparent groups.
Adjusted Rand index	Agreement between two groupings; 1 identical, 0 chance.

B Sources of the results

Each number in this report was read from saved result files. A sheet listing every quoted value next to its source is produced by `report/figures/make_post_midterm_sources.py`. The main sources are: Gaussian process ratio forecasting (`outputs/step3_gpr/`); the joint mixed model (`outputs/joint_subscore/`); the held-out comparison of subscore models, the HMM stability study and the temporal analysis (cluster-phase experiment runs, summarised in `docs/cluster_phase/04_RESULTS_GENERATED.md`); the clustering audit (`outputs/clustering_audit/`) and its closure (`docs/cluster_phase/05_FINDINGS.md`, with the specifications and run records listed there).